

Confluent Kafka Isotope

An example of using Kafka **consumer and producer interceptors** to tag every message with a tracer — an *isotope* — that carries through every hop of a multi-topic pipeline. Apache Flink consumes the tagged records and reports on what the isotopes reveal: **end-to-end latency**, **hop topology**, **drop/duplication rates**, and **pipeline coverage**.

A **portability requirement** runs through this project: the same isotope mechanism must work against both **Confluent Cloud for Apache Flink (CCAF)** (managed) and **Confluent Platform for Apache Flink** (self-managed) — both used here via Table API SQL plus uploaded UDF/PTF JARs (no DataStream code on either side). Three decisions follow from that: tagging happens in the Kafka **producer/consumer interceptors** (the one extension point both runtimes share via the broker), the on-wire **header** format is **JSON** (so Flink SQL can read the scalar fields with `CAST(headers [...] AS STRING)` and no UDF), and the optional stateful reports (`LatencyPercentilesUDAF`, `StuckTracePTF`) ship as a single JAR that registers identically on either runtime.

One asymmetry the runtimes don't share: **Flink-native SR-Protobuf**. CCAF supports SR-framed Protobuf as a Flink sink format via its topic catalog; Apache Flink open-source (the CP Flink runtime) ships `avro-confluent` but no SR-Protobuf counterpart. So Flink *report sinks* land on **Avro+SR on CP** and can be **Protobuf+SR on CCAF** — a runtime constraint, not a project preference. The demo *event* topics (next paragraph) are unaffected because they're written by the Kafka producer client, not by Flink.

Message **values** on the demo topics are **SR-framed Protobuf** (`ai.signalroom.kafka.isotope.proto.DemoEvent`) — the standard Confluent value format. The interceptors and reports are agnostic to value format because the isotope rides in headers; the Protobuf choice just gives the integration tests and the demo CLI a typed payload to work with.

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1.0 How the isotope is carried

- **Header `x-isotope`** (JSON bytes) carries the full hop history, forwarded by every hop:
 - `t` — 16-byte **UUIDv7** trace ID (RFC 9562): 48-bit ms timestamp in the high bits + 74 bits random. Stable for the life of the trace, and lexicographic byte order matches creation order

- sort trace IDs and you get chronological order for free.
- **o** — origin timestamp (ms) — same value as the timestamp embedded in the UUIDv7 trace ID; kept as its own field for typed access from Flink SQL without needing to decode the UUID bytes.
- **s** — origin service name (set once, never reassigned)
- **h** — ordered list of hops, each `{s: service, t: topic, m: tsMs}`
- **x** — **true** if the hop list exceeded `MAX_HOPS = 32` and the oldest hop was evicted
- **Six scalar headers** (UTF-8 strings) carry the most-recent-hop view so Flink SQL can read them via `CAST(headers['x-isotope-...'] AS STRING)` without parsing the JSON array (no UDF required on either CCAF or CP Flink). See <link/README.md> for the full header table.

A producer with the isotope interceptor loaded appends one hop on every `send()`. A consume-then-produce service calls `IsotopeContext.adoptFromRecord(record)` between consume and produce so the trace ID and origin survive the hop.

2.0 Repo layout

app/	isotope JVM library + demo CLI +
tests	
src/main/proto/	DemoEvent.proto (Protobuf message
value)	
src/main/java/ai/signalroom/kafka/isotope/	
Isotope.java	POJO + JSON codec + Hop +
fromHeaders()	
	+ UUIDv7 helpers (uuidV7Bytes /
uuidV7String)	
IsotopeContext.java	ThreadLocal + adoptFromRecord()
IsotopeProducerInterceptor.java	stamps/appends x-isotope + 6 scalar
	reporting headers on send()
IsotopeConsumerInterceptor.java	batch-aware logging; no auto-
propagation	
App.java	demo CLI — send / hop / sink modes
src/test/java/.../	IsotopeCodecTest (no broker needed)
src/integrationTest/java/.../	live-broker tests; produce/consume
	DemoEvent via SR-framed Protobuf
	(need Minikube CP + SR port-
forwarded)	
ptf/	Phase 2 — Flink PTF + UDAF shadow
JAR	
src/main/java/ai/signalroom/kafka/isotope/flink/	
LatencyPercentilesUDAF.java	T-Digest p50/p95/p99 aggregate
StuckTracePTF.java	per-trace state + event-time timer
Percentiles.java, StuckTraceAlert.java	return-type DT0s
src/test/java/.../	LatencyPercentilesUDAFTest
flink/sql/	Flink SQL reports — identical on
	CCAF and CP Flink except source DDL
cp/	CP Flink: 00_source_table,
01_register_functions,	
	05_isotope_view, 05_report_sinks
(avro-confluent),	
	10/20/30/40/60/70 INSERT INTO

```
reports, 99_tearardown
(CCAF SQL is inlined under

terraform/setup-confluent-flink.tf;
CP SQL is hardcoded in

flink/sql/cp/.)
CFK Kafka/SR/Connect/ksqlDB/C3
k8s/base/
manifests
scripts/
port-forward helpers, deploy-cp-
flink-reports.sh,
deploy-cc-flink-reports.sh
Makefile
cp-up / flink-up / kafka-pf-up /
...

```

3.0 Running

3.1. Unit tests (no broker, instant)

```
./gradlew test # both subprojects
# or scoped:
./gradlew :app:test # IsotopeCodecTest - JSON
roundtrip, hop eviction, header size, UUIDv7 properties (10 tests)
./gradlew :ptf:test # LatencyPercentilesUDAFTest - T-
Digest accumulator semantics

```

3.2 Demo CLI — see one trace propagate live

The fastest way to watch the isotope mechanic. Requires the cluster to be up and the Kafka + SR forwards running (see step 3 below for the bring-up commands). The CLI has three modes:

Mode	Args	What it does
send	<topic> <service> <payload>	Produces one isotope-tagged DemoEvent to <topic>, then exits. Auto-creates the topic.
hop	<in-topic> <out- topic> <service>	Consumes records from <in-topic>, adopts the isotope into thread-local, re-produces the same DemoEvent to <out-topic> as <service> (which appends a new hop). Runs until Ctrl-C.
sink	<topic>	Subscribes to <topic> and pretty-prints the full isotope trail for every arriving record. Runs until Ctrl-C.

A 3-stage chain in four terminals:

```
# Terminal A - terminal sink (will print the full 3-hop trail)
./gradlew :app:run --args="sink iso_final" -q

# Terminal B - middle stage: iso_mid -> iso_final as svc-C

```

```
./gradlew :app:run --args="hop iso_mid iso_final svc-C" -q

# Terminal C – first stage: iso_start → iso_mid as svc-B
./gradlew :app:run --args="hop iso_start iso_mid svc-B" -q

# Terminal D – kick the chain off (run repeatedly to send more)
./gradlew :app:run --args="send iso_start svc-A 'hello world'" -q
```

Terminal A's output for each record shows the same `trace_id` across all three hops, `origin = svc-A` (never reassigned), and `hops[]` listing `svc-A → svc-B → svc-C` in order with per-hop timestamps. Override endpoints via `-Dkafka.bootstrap=...` / `-Dschema.registry.url=...` if you're not on the default Minikube layout.

3.3 Integration tests (live Kafka via Minikube)

Bring up the local Confluent Platform stack and port-forward Kafka + SR:

```
make minikube-start          # one-time
make cp-up                  # CFK Operator +
Kafka/SR/Connect/ksqlDB/C3 (~5 min)
make kafka-pf-up           # localhost:30092 → Kafka,
                           localhost:8081 → Schema Registry
```

Then run the suite:

```
./gradlew :app:integrationTest          #
all 5 tests
./gradlew :app:integrationTest --tests '*ProducerInterceptorIT'  #
just one
```

Override the endpoints if needed:

```
./gradlew :app:integrationTest \
  -PkafkaBootstrap=localhost:30092 \
  -PschemaRegistryUrl=http://localhost:8081
```

Tear down forwards when done:

```
make kafka-pf-down
```

The integration tests cover:

Test	What it verifies
<code>BrokerSmokeIT</code>	<code>AdminClient</code> can create/list/delete a topic via the <code>NodePort</code> port-forward
<code>ProducerInterceptorIT</code>	A bare consumer sees the <code>x-isotope</code> JSON header + all 6 scalar reporting headers with the expected origin/hop values, and the Protobuf round-trip preserves <code>DemoEvent.source / payload</code>
<code>ConsumerInterceptorIT</code>	<code>IsotopeContext.adoptFromRecord</code> extracts isotope into thread-local on tagged records; clears the thread-local for untagged records
<code>ThreeStageHopPropagationIT</code>	<code>svc-A → topic-AB → svc-B → topic-BC → svc-C</code> produces a stable trace ID, 2-hop trail in send order, and correct scalar headers (origin = <code>svc-A</code> , this = <code>svc-B</code> , hop count = 2) at the terminal

3.4 Flink SQL reports on Minikube

The four Phase-1 reports plus the Phase-2 PTF and UDAF reports — six in total — run against a Flink session cluster managed by the Confluent Flink Kubernetes Operator. Same FQL files deploy to Confluent Cloud for Apache Flink — see § 3.5 for that path; this section is the local-Minikube path.

Bring up Flink:

```
make flink-up          # cert-manager → CFK Flink Operator → CMF →
session cluster       # (~5 min the first time)
```

Deploy the reports — all 6 reports run on the `cp-flink` session cluster (Flink 2.1.2). Sink topics use Apache Flink's `avro-confluent` format — SR-framed Avro, auto-registered on first write — so Control Center renders the report rows natively.

```
make flink-reports-up
```

`flink-reports-up` builds the PTF/UDAF shadow JAR if missing, copies it into the JobManager pod, pre-creates the 6 sink Kafka topics, then applies the source + view + sink DDL and submits 6 `INSERT INTO` streaming jobs (one per report). On first write to each sink, Apache Flink's `flink-sql-avro-confluent-registry` format registers a fresh Avro schema in SR under subject `<topic>-value`. **Control Center deserializes all 6 report topics natively** — no `.proto` files in the repo for the reports, no hand-installed format jars beyond the one init-container download.

3.4.1 Format-by-domain

The demo event topics (`iso_start`, `iso_mid`, `iso_final`) still ride **Protobuf+SR** via the Java app's `DemoEvent` schema — that's unchanged. The `report` topics ride **Avro+SR** because cp-flink doesn't ship an SR-integrated Protobuf format and CMF (which does) disallows the UDAFs the percentiles report needs. Events from the app are Protobuf; aggregates from Flink are Avro. Two formats by domain — a clean split, not a defect.

3.5 Flink SQL reports on Confluent Cloud for Apache Flink (CCAF)

CCAF parallel of § 3.4, driven by Terraform under [terraform/](#). One `make` target spins up a fresh Confluent Cloud environment, Kafka cluster, 9 topics (3 isotope event + 6 report sinks), a Flink compute pool, a rotating service-account API key pair, the PTF/UDAF JAR uploaded as a Flink artifact, and 16 long-lived `confluent_flink_statement` resources (source view, typed view, 6 sinks, 2 `CREATE FUNCTION`, 6 streaming `INSERT INTO`). The Terraform shape mirrors `apache_flink-kickstarter-ii` — same provider version, same `iac-confluent-api_key_rotation-tf_module`, same DROP-then-CREATE statement pattern.

Prereqs:

- [Terraform](#) `>= 1.13` installed locally.
- A Confluent Cloud API key (Cloud-level, not cluster-scoped) with permissions to create environments, Kafka clusters, Flink compute pools, service accounts, role bindings, Flink artifacts, and statements. Generate via Console → Settings → Cloud API keys.

Deploy:

```
export CONFLUENT_API_KEY=...
export CONFLUENT_API_SECRET=...
make cc-flink-reports-up CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

The wrapper script ([scripts/deploy-cc-flink-reports.sh](#)) builds the PTF/UDAF shadow JAR if missing, then runs `terraform apply -auto-approve` in [terraform/](#). First-run takes ~6–8 minutes (Kafka cluster provisioning dominates). Re-applies are idempotent — `CREATE ... IF NOT EXISTS` plus `lifecycle { ignore_changes = [compute_pool] }` on every statement.

What gets created (see [terraform/setup-confluent-flink.tf](#) for the full graph):

Resource	Name	Notes
<code>confluent_environment</code>	<code>confluent-kafka-isotope</code>	ESSENTIALS stream-governance package
<code>confluent_kafka_cluster</code>	<code>kafka-isotope</code>	Standard, single-zone, AWS us-east-1 by default
<code>confluent_kafka_topic</code> × 9	<code>iso-{start,mid,final}, isotope-report-*1m</code>	Explicit so <code>destroy</code> cleans them up

Resource	Name	Notes
<code>confluent_service_account</code> + 5 role bindings	<code>isotope-flink-sql-runner</code>	FlinkDeveloper + ResourceOwner-topic + Assigner + SR-subject + transactional
<code>confluent_flink_compute_pool</code>	<code>isotope-flink-statement-runner</code>	10 CFU; comfortable headroom for 6 INSERTs + ad-hoc SELECTs
<code>confluent_flink_artifact</code>	<code>isotope-flink-udf</code>	Uploads <code>ptf/build/libs/isotope-flink-udf.jar</code>
<code>confluent_flink_statement</code> × 16	(see file)	View + typed view + 6 sinks + 2 functions + 6 INSERTs

Useful outputs:

```

terraform -chdir=terraform output environment_id
terraform -chdir=terraform output kafka_bootstrap_servers
terraform -chdir=terraform output schema_registry_url
terraform -chdir=terraform output -raw kafka_api_key      # sensitive
terraform -chdir=terraform output -raw kafka_api_secret  # sensitive

```

Format-by-runtime (not -by-domain). CP's reports land on **Avro+SR** (`'value.format' = 'avro-confluent'` in `flink/sql/cp/05_report_sinks.fql`). CCAF's reports land on **Protobuf+SR** (`'value.format' = 'proto-registry'` in each sink's WITH clause in `terraform/setup-confluent-flink.tf`). The two runtimes' SQL is otherwise unshared: CP's lives hardcoded in `flink/sql/cp/`, CCAF's lives inline as `confluent_flink_statement` resources in `terraform/setup-confluent-flink.tf`.

CCAF UDAF limitation. CCAF currently rejects all `CREATE FUNCTION` statements for user-defined aggregate functions ("aggregate functions are not supported"). The `LATENCY_PERCENTILES` UDAF — Phase-2 of the project — therefore deploys on CP only; the percentile report (`latency_percentiles_flat_1m`) does not exist on CCAF. The other Phase-2 function, `STUCK_TRACE_PTF` (a `ProcessTableFunction`, not a UDAF), works on both runtimes. The JAR itself is portable — `LatencyPercentilesUDAF` ships with a `byte[]`-based accumulator and is ready to register when Confluent lifts the restriction. The five CCAF reports are: `latency` (avg/min/max), `topology`, `hop_distribution`, `coverage`, `stuck_trace`.

Driving traffic — the 3-stage demo against CCAF. `App.java` reads four optional `-D` properties (`kafka.security.protocol`, `kafka.sasl.mechanism`, `kafka.sasl.jaas.config`, `schema.registry.basic.auth.user.info`) that default to plaintext-no-auth for Minikube. `scripts/cc-cli-env.sh` pulls the Kafka + Schema-Registry credentials from `terraform output` (both keys are rotated by `module.kafka_api_key_rotation` and `module.sr_api_key_rotation` in `terraform/setup-confluent-kafka.tf`) and builds the JAAS string.

The thin wrapper `scripts/cc-app-run.sh` sources the env helper then invokes `./gradlew :app:run` with the six `-D` flags — pass it the same `send` / `hop` / `sink` args you'd give `App.java`:


```
# Four terminals (same A/B/C/D order as § 3.2). No manual env exports –
# the wrapper sources cc-cli-env.sh, which pulls everything from terraform.
scripts/cc-app-run.sh sink iso_final # A
scripts/cc-app-run.sh hop iso_mid iso_final svc-C # B
scripts/cc-app-run.sh hop iso_start iso_mid svc-B # C
scripts/cc-app-run.sh send iso_start svc-A 'hello' # D
```

The wrapper hard-fails with a clear message if any of the seven required values is missing, so you'll never silently hand gradle empty `-D` values.

Terminal A prints the same `trace_id` across all three hops, and the CCAF report INSERTs populate as you fire Terminal D — `SELECT * FROM isotope_report_latency_1m` etc. in the Cloud Console SQL workspace.

Sustained traffic — required to see report rows. The five INSERT INTO jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, and a tumbling window only emits when the watermark advances past `window_end`. A handful of records bursted from Terminal D within a single 1-minute interval will sit in one open window forever (the most-recent record is the watermark, and it never gets older than itself). Spread traffic across **multiple** windows so the watermark crosses each boundary:

```
# 30 records spaced 5s apart ≈ 2.5 minutes of event-time → spans 3+ windows
for i in {1..30}; do
  scripts/cc-app-run.sh send iso_start svc-A "burst-$i"
  sleep 5
done
```

Wait ~90 seconds after the *last* record before checking `isotope_report_latency_1m` (and friends) — that's the watermark catching up. The `stuck_trace_alerts_1m` sink only fires for traces that go ≥60s of event time without a fresh hop, so the burst above won't trigger it (every trace gets one record and ends — no stalled in-flight state). To exercise `STUCK_TRACE_PTF`: send a single record to `iso_start` and don't run the `svc-B` / `svc-C` hops, then keep sending unrelated records elsewhere so the watermark advances past the stuck trace's `event_time + 60s`.

Tip — auto-source after apply. `make cc-flink-reports-up` runs in its own subshell, so it can't export env vars back into yours. Add this to your `~/.zshrc` / `~/.bashrc` for a one-liner that applies and exports:

```
cc-up() {
  make cc-flink-reports-up "$@" && source scripts/cc-cli-env.sh
}
cc-down() {
  make cc-flink-reports-down "$@" && unset BOOTSTRAP SR_URL KAFKA_KEY
  KAFKA_SECRET SR_KEY SR_SECRET JAAS
}
# then: cc-up CONFLUENT_API_KEY=... CONFLUENT_API_SECRET=...
```


Teardown:

```
make cc-flink-reports-down CONFLUENT_API_KEY=$CONFLUENT_API_KEY  
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

Runs `terraform destroy -auto-approve` — deletes every resource above, including the environment itself. Safe to run repeatedly.

3.6 Recommended path the first time through

1. `./gradlew test` — proves the codec + UDAF logic without any cluster.
2. `make cp-up && make kafka-pf-up && ./gradlew :app:integrationTest` — proves the broker + SR + interceptor + Protobuf path end-to-end.
3. The 3-stage demo CLI walkthrough above — visually shows the trace accumulating hops.
4. `make flink-up && make flink-reports-up && make flink-sql` — reports populate as you drive traffic via the demo CLI (see § 3.2).
5. (Optional) `make cc-flink-reports-up` — the CCAF parallel; see § 3.5 for prereqs and the SASL-config caveat for the demo CLI.